

**DEPARTMENT OF COMPUTER ENGINEERING
CSD415 MAJOR PROJECT**

Academic Year: 2026-2027

Class: S7 C

Project Approval Form

Date:

Roll No	KTU Reg No	Name	Signature
17	CHN23CS041	ANANYA P SANTH	
42	CHN23CS121	L S POOJA LEKSHMI	
50	CHN23CS143	NEHA JOSE	
61	CHN23CS190	T A DEVINANDANA	

Submit three project topic proposals with a very small description. At the end of each topic mention two major references of the topic you referred. Refer international journals of repute like IEEE / Elsevier / Springer/ other SCI indexed journals for selecting seminar topic. (Give the topics in your preference order)

Name of Topic 1: HERBALANCE AI

Abstract: HERBALANCE AI is an AI powered Women's Health and Wellness Companion that integrates beauty, health, and wellness into a single intelligent platform. Women often rely on multiple applications for skincare, medication management, menstrual tracking, and fitness, making personalized healthcare fragmented and inconvenient. The platform analyzes users' skin type and skin tone to recommend suitable cosmetic products and identify harmful ingredients. It also provides medicine checking, AI based menstrual cycle prediction, secure health record management, and personalized workout and wellness recommendations to support conditions such as PCOS and PCOD. By combining these features in one user friendly application, the system enables women to make informed decisions, improve self care, and maintain a healthier lifestyle through personalized AI driven insights.

References:

[1] S. A. Alkhodair, A. Alshammari, and H. Alqahtani, "Women's Health Recommendation System Using Large Language Models," *IEEE IATMSI*, 2025.

[2] J. Liu et al., "Machine Learning Based Menstrual Phase Identification Using Wearable Device Data," *npj Women's Health*, vol. 2, 2025.

Name of Topic 2: MediConnect: Intelligent Symptom Prediction and Appointment System

Abstract: Healthcare systems often require patients to manually identify the appropriate medical department, contact hospitals, and schedule appointments, making the process time-consuming and inconvenient. This project proposes an AI-Based Hospital Management System that integrates a Machine Learning disease prediction model with an intelligent chatbot to simplify hospital services. The chatbot interacts with users by collecting symptoms in natural language, predicts the most probable disease using a trained machine learning model, and recommends the appropriate medical department and specialist doctor. Patients can then directly book appointments based on doctor availability and receive instant confirmation along with a downloadable health report and email notification. The system also provides a complete hospital management platform where administrators can manage doctors, departments, appointments, patient records, and availability schedules through a secure dashboard. Machine Learning algorithms such as Random Forest are used for symptom-based disease prediction, while Natural Language Processing (NLP) enables conversational interaction between the patient and the chatbot. The proposed system improves accessibility, reduces hospital waiting time, assists patients in choosing the correct specialist, and enhances the overall efficiency of healthcare services. The solution is designed using open-source technologies, making it cost-effective, scalable, and suitable for hospitals, clinics, and educational research projects.

References:

[1] Sanjay Chakraborty, Hrithik Paul, Sayani Ghatak, et al., "An AI-Based Medical Chatbot Model for Infectious Disease Prediction," *IEEE Access*, vol. 11, pp. 128469–128484, 2023.

[2] Subhasish Mohapatra and Abhishek Roy, "Analysing Weighted Disease Symptoms to Recommend an Expert Through an Integrated Healthcare Model," *Discover Public Health*, vol. 21, Article 130, 2024.

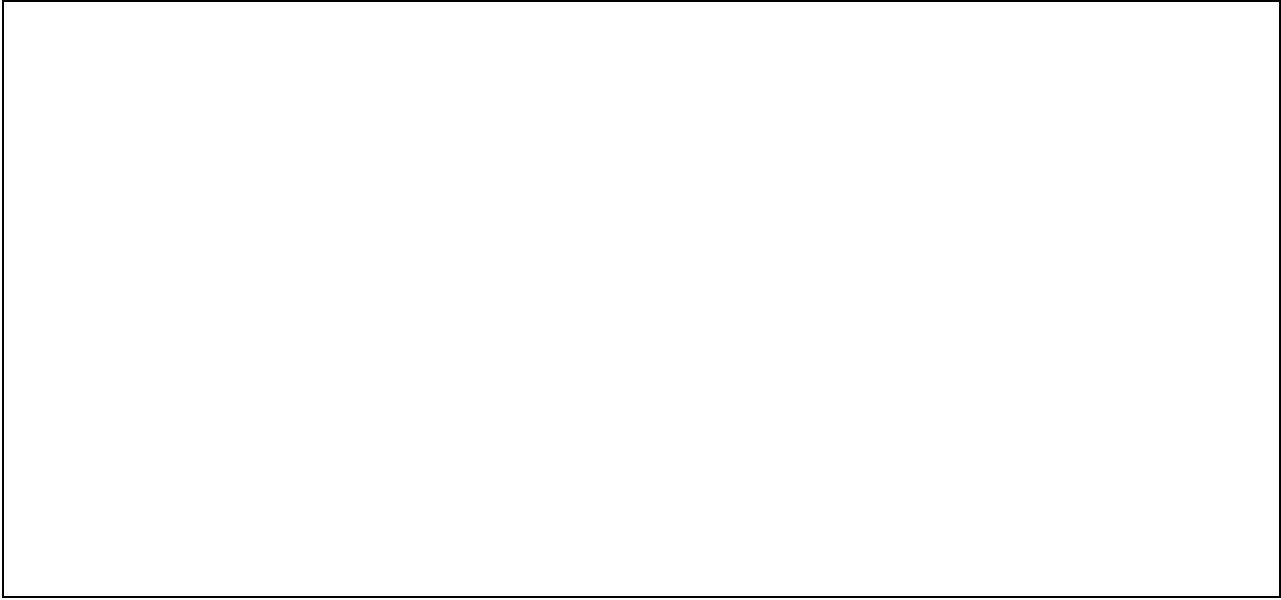
Name of Topic 3: NEXT11:AI-Powered Football Talent Discovery and Grassroots Development Platform

Abstract: NEXT11 is an AI-powered Football Talent Discovery and Grassroots Development Platform designed to strengthen football development in Kerala by connecting players, coaches, academies, and tournament organizers through a unified digital platform. It addresses the challenges of identifying talented players, especially from rural areas, where opportunities for professional scouting and performance evaluation are limited. The platform leverages Artificial Intelligence (AI) and Computer Vision to analyze football players' performance from match or training videos. Using technologies such as OpenCV, MediaPipe, YOLO, and TensorFlow, the system evaluates key performance metrics including dribbling, passing, shooting, speed, movement, and positioning. Based on the analysis, it generates personalized performance reports and improvement suggestions, enabling coaches and academies to identify and recruit promising players more effectively. In addition to AI-based talent discovery, the platform offers features such as tournament management, player progress tracking, football ground booking, and injury-risk prediction, creating a comprehensive digital ecosystem for grassroots football. By integrating Flutter, Python, FastAPI, and Firebase, the proposed solution aims to make football talent identification more transparent, accessible, and data-driven while promoting equal opportunities for aspiring players and supporting the long-term growth of football in Kerala.

References:

[1] B. M. Merzah et al., "Football Player Tracking and Performance Analysis Using the OpenCV Library," *Mathematical Modelling of Engineering Problems*, vol. 11, no. 1, 2024.

[2] S. Sneha et al., "Machine Learning Based Assessment of Elite Football Players Based on Anthropometric and Motor Fitness Parameters," *SN Computer Science*, 2024.



Guide's Name :

Guide's Suggestion (if any):

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Programme Assessment Committee (PAC):

1. Approved/Rejected
2. Approved/Rejected
3. Approved/Rejected
Selected Topic:

Project Co-Ordinator

HOD